

Lab 7 - Harry Mendez

$S \rightarrow OAO | 1B1 | BB$

$A \rightarrow C$

$B \rightarrow S | A$

$C \rightarrow S | \epsilon$

Paso 1

$S \rightarrow OAO | 1B1 | BB | OAO$

$S \rightarrow OAO | 11 | 1B1 | B$

$S \rightarrow OAO | OO | 1B1 | 11 | BB | B$

$S \rightarrow OAO | OO | 1B1 | 11 | BB | B$

$A \rightarrow C$

$B \rightarrow S | A$

$C \rightarrow S$

Paso 2

$P_0 = \{(S,S), (A,A), (B,B), (C,C)\}$

$(S,S) \quad S \rightarrow B \quad (S,B)$

$(S,B) \quad B \rightarrow A \quad (S,A)$

$(S,A) \quad A \rightarrow \epsilon \quad (S,C)$

$(S,C) \quad C \rightarrow S \quad (S,S)$

$(B,B) \quad B \rightarrow A \quad (B,A)$

$(B,A) \quad A \rightarrow C \quad (B,C)$

$(B,C) \quad C \rightarrow S \quad (B,S)$

$(A,A) \quad A \rightarrow C \quad (A,C)$

$(A,C) \quad C \rightarrow S \quad (A,S)$

$(C,C) \quad C \rightarrow S \quad (C,S)$

$(S,S) \quad S \rightarrow OAO | 1B1 | OO | 11 | BB$

$(B,S) \quad S \rightarrow OAO | 1B1 | OO | 11 | BB$

$(A,S) \quad S \rightarrow OAO | 1B1 | OO | 11 | BB$

$(C,S) \quad S \rightarrow OAO | 1B1 | OO | 11 | BB$

$S \rightarrow OAO | 1B1 | OO | 11 | BB$

$A \rightarrow OAO | 1B1 | OO | 11 | BB$

$B \rightarrow OAO | 1B1 | OO | 11 | BB$

$C \rightarrow OAO | 1B1 | OO | 11 | BB$

Paso 3

$T = \{O, 1\} \quad \omega_1 = \{S, A, B, C\} \quad \omega_2 = \{S, A, B, C\}$

$y_1 = \{S\}$

$y_2 = \{S, A, B, C, 1\}$

$y_2 = \{S, A, B, C, 1\}$

$S \rightarrow OAO | 1B1 | OO | 11 | BB$

$A \rightarrow OAO | 1B1 | OO | 11 | BB$

$B \rightarrow OAO | 1B1 | OO | 11 | BB$

Forma Chomsky:

$r_0 = O \quad r_1 = 1$

$S \rightarrow T_0 A T_0 | T_1 B T_1 | O O | 1 1 | B B$

$A \rightarrow T_0 A T_0 | T_1 B T_1 | O O | 1 1 | B B$

$B \rightarrow T_0 A T_0 | T_1 B T_1 | O O | 1 1 | B B$

$C_1 = A T_0 \quad C_2 = B T_1$

$S \rightarrow T_0 C_1 | T_1 C_2 | O O | 1 1 | B B$

$A \rightarrow T_0 C_1 | T_1 C_2 | O O | 1 1 | B B$

$B \rightarrow T_0 C_1 | T_1 C_2 | O O | 1 1 | B B$

2. $S \rightarrow aAa \mid bBb \mid \epsilon$
 $A \rightarrow C \mid a$
 $B \rightarrow C \mid b$
 $C \rightarrow CDE \mid C$
 $D \rightarrow A \mid B \mid ab$

Paso 1

$S \rightarrow aAa \mid aa \mid bBb$
 $S \rightarrow aAa \mid bb \mid bBb$
 $S \rightarrow aAa \mid aa \mid bb \mid bBb$
 $A \rightarrow C \mid a$
 $B \rightarrow C \mid b$
 $C \rightarrow CDE \mid DE \mid CE \mid E$
 $D \rightarrow A \mid B \mid ab$

Paso 2

$P_0 = \{(S,S), (A,A), (B,B), (C,C), (D,D)\}$
 $(C,C) \ C \rightarrow E \ (C,E)$

$(A,A) \ A \rightarrow C \ (A,C)$
 $(A,C) \ C \rightarrow E \ (A,E)$

$(B,B) \ B \rightarrow C \ (B,C)$
 $(B,C) \ C \rightarrow E \ (B,E)$

$(D,D) \ D \rightarrow A \ (D,A)$

$(D,A) \ A \rightarrow C \ (D,C)$

$(D,C) \ C \rightarrow E \ (D,E)$

$(D,D) \ D \rightarrow B \ (D,B)$

(D,B)

$(C,C): CDE \mid DE \mid CE$

$(C,E):$ No produce nada

$(S,S): aAa \mid aa \mid bb \mid bBb$

$(A,A): C$

$(A,C): CDE \mid DE \mid CE$

$(B,B): b$

$(B,C): CDE \mid DE \mid CE$

$(D,D): ab$

$(D,A): a$

$(D,B): b$

$(D,C): CDE \mid DE \mid CE$

$(D,E):$ No produce nada

$S \rightarrow aAa \mid aa \mid bb \mid bBb$

$A \rightarrow CDE \mid DE \mid CE \mid a$

$B \rightarrow CDE \mid DE \mid CE \mid b$

$C \rightarrow CDE \mid DE \mid CE$

$D \rightarrow CDE \mid DE \mid CE \mid ab \mid a \mid b$

Poso 3

$T: \{a, b\}$ $V_1: \{S, A, B, D, a, b\}$ $W_2: \{S, A, B, C, D, a, b\}$
 $W_3: \{S, A, B, C, D, a, b\}$

$y_1 \rightarrow \{S\}$ $y_2 \rightarrow \{D, A, B, a, b, S\}$ $y_3 \rightarrow \{D, A, B, a, b, C, S\}$

$S \rightarrow \cancel{a}Aa \mid aa \mid b\cancel{b} \mid bBb$
 $A \rightarrow CDE \mid DE \mid CE \mid a$
 $B \rightarrow CDE \mid DE \mid CE \mid b$
 $C \rightarrow CDE \mid DE \mid CE$
 $D \rightarrow CDE \mid DE \mid CE \mid ab \mid a \mid b$

Forma Chomsky:

$C_1 = DE$ $T_1 = C$ $T_2 = b$

$S \rightarrow T_1AT_1 \mid T_1T_1 \mid T_2T_2 \mid T_2BT_2$
 $A \rightarrow CC_1 \mid DE \mid CE \mid T_1$
 $B \rightarrow CC_1 \mid DE \mid CE \mid T_2$
 $C \rightarrow CC_1 \mid DE \mid CE$
 $D \rightarrow CC_1 \mid DE \mid CE \mid T_1T_2 \mid T_1 \mid T_2$

$C_2 = AT_1$ $C_3 = BT_2$

$S \rightarrow T_1C_2 \mid T_1T_1 \mid T_2T_2 \mid T_2C_3$
 $A \rightarrow CC_1 \mid DE \mid CE \mid T_1$
 $B \rightarrow CC_1 \mid DE \mid CE \mid T_2$
 $C \rightarrow CC_1 \mid DE \mid CE$
 $D \rightarrow CC_1 \mid DE \mid CE \mid (T_1T_2)T_1 \mid T_2$

3. $S \rightarrow ASA | aB$
 $A \rightarrow B | S$
 $B \rightarrow b | c$

Paso 1
 $S \rightarrow ASA | c | aB$
 $S \rightarrow SA | AS | ASA | cB$
 $S \rightarrow ASA | SA | AS | aB | a$
 $S \rightarrow ASA | SA | AS | aB | a | S$
 $A \rightarrow B | S$
 $B \rightarrow b$

Paso 2
 $S \rightarrow A \rightarrow B$
 $P_0 = \{(S, S), (A, A), (B, B)\}$

$(A, A) A \rightarrow B (A, B)$
 $(A, B) A \rightarrow S (A, S)$
 (A, A) : No abierta
 (A, S) : $ASA | AS | SA | cB | a$
 (A, B) : b

$S \rightarrow ASA | AS | SA | aB | a$
 $A \rightarrow ASA | AS | SA | aB | c | b$
 $B \rightarrow b$

Paso 3 $T: \{a, b\}$ $W_1: \{S, A, B, a, b\}$ $W_2: \{S, A, B, a, b\}$

$y_1: \{S\}$
 $y_2: \{S, A, a, b\}$
 $y_3: \{S, A, B, a, b\}$
 $y_4: \{S, A, B, a, b\}$

$S \rightarrow ASA | AS | SA | aB | a$
 $A \rightarrow ASA | AS | SA | aB | c | b$
 $B \rightarrow b$

Forma Chomsky:

$C_1 = AS$ $T_1 = a$ $T_2 = b$
 $S \rightarrow C_1 A | AS | SA | T_1 B | T_1$
 $A \rightarrow C_1 A | AS | SA | T_1 B | T_1 | T_2$

$B \rightarrow T_2$